Kaarthik M
Pre-Final Year Student | Student Placement Coordinator | 5x Patent Published | 4x Hackathon Winner | Developer | Radio Jockey

Patents

AI-Based Predictive Thermal Monitoring and Adaptive Charging Optimization System for Preventing Battery Swelling in Consumer Smartphones and Laptops

202641110142 · Filed Sep 18, 2026

Published patent application for an AI-based, on-device battery safety framework designed to predict thermal stress and battery swelling risk in consumer smartphones and laptops. The system integrates Edge AI-based swelling prediction, Thermal Stress Score and Swelling Probability Score generation, micro thermal gradient analysis, behavioral charging learning, application-level heat mapping, CPU/GPU-aware charging control, battery age compensation, adaptive charging orchestration, and self-learning optimization. The framework dynamically adjusts charging behavior based on real-time thermal conditions, battery health, workload, and usage patterns while operating without cloud connectivity or additional external hardware.

System and Method for Dynamic Cyber DNA Fingerprinting and Evolutionary Analysis of Cybercrime Complaints

202641108530 · Filed Sep 18, 2026

Published Patent Application for a system and method that introduces Dynamic Cyber DNA Fingerprinting and Evolutionary Analysis for cybercrime complaints. The invention processes multimodal digital evidence and represents cybercrime incidents through interconnected behavioral and intent-based Cyber DNA genes. It enables correlation of seemingly unrelated complaints, detection of behavioral mutations, construction of Cyber DNA evolutionary lineages, and explainable intelligence to support cybercrime investigation.

A System and Method for Tokenless Laundry Bag Identity Generation, Localization, Verification and Recovery

202641104992 · Filed Sep 11, 2026

Published patent application for an AI-based tokenless laundry bag management system that enables laundry bag identity generation, autonomous localization, verification, and recovery without relying on physical identifiers such as RFID tags, QR codes, or tokens. The system uses computer vision and AI to generate a unique Laundry Identity Profile based on visual and physical characteristics such as color, shape, texture, dimensions, appearance, and approximate weight. It enables autonomous rack localization, collection verification, and misplaced bag recovery through similarity matching and AI-based re-identification.

Adaptive Capability Reallocation Framework for Intelligent Railway Operational Safety

202641101765 · Filed Aug 28, 2026

A computer-implemented framework for intelligent railway operational safety that enables adaptive reallocation of individual railway operational capabilities among intelligent agents during partial system degradation. The framework integrates Railway Digital Twin technology, multi-agent coordination, Capability Health Assessment, predictive degradation detection, Capability Reallocation Scoring, railway-specific safety validation, dependency management, and automatic rollback and recovery. Unlike conventional systems that replace an entire failed agent, the proposed approach selectively reallocates only the degraded capability while maintaining unaffected functions, thereby improving operational continuity, fault tolerance, resource utilization, resilience, and safety in intelligent railway signalling, electronic interlocking, train movement supervision, and railway traffic management systems.

System For Predictive Power-Aware Adaptive Neural Inference

202641065402 · Filed May 24, 2026

Patent application for a predictive power-aware adaptive neural inference system that dynamically adjusts neural network configurations based on forecasted power conditions, enabling energy-efficient and uninterrupted AI execution in edge computing environments.

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